Assingment —2

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Task 1: Statistical Summary for Survey Data

Prompt: generate a Python function that reads a list of numbers and calculates the mean, minimum, and maximum values. Code:

```
def calculate_statistics(numbers):    if
not numbers:
    return None, None, None  # Return None for mean, min, and max if the list is empty


  mean_value = sum(numbers) / len(numbers)
min_value = min(numbers)    max_value =
max(numbers)


  return mean_value, min_value, max_value  #exapmple
usage:
numbers = [10, 20, 30, 40, 50] mean, minimum, maximum =
calculate_statistics(numbers) print(f"Mean: {mean}, Minimum:
{minimum}, Maximum: {maximum}")
```

output

:



Task 2: Armstrong Number – AI Comparison

Prompt: Generate an Armstrong number checker using Given a number, check if it is an Armstrong number or not. Compare their outputs, logic style, and clarity.

Code:

```
def is_armstrong_number(n):
    # Convert the number to a string to easily iterate over each digit
digits = str(n)     num_digits = len(digits)

    # Calculate the sum of each digit raised to the power of the number of digits
armstrong_sum = sum(int(digit) ** num_digits for digit in digits)

    # Check if the calculated sum is equal to the original number
return armstrong_sum == n # Example usage: number = 153     if
is_armstrong_number(number):
    print(f"{number} is an Armstrong number.") else:
    print(f"{number} is not an Armstrong number.")
```

output:



Task 3: Leap Year Validation Using Cursor AI

Prompt: generate a Python program that checks whether a given year is a leap year.also give two different methods to check for leap year and compare their outputs, logic style, and clarity.

Code:

```
# Method 1: Using the standard leap year rules def
is_leap_year_method1(year):     if (year % 4 == 0 and year %
100 != 0) or (year % 400 == 0):
        return True
else:
        return False
# Example usage:
year = 2020  if
is_leap_year_method1(year):
print(f"{year} is a leap year.") else:
print(f"{year} is not a leap year.")


# Method 2: Using a more concise approach def
is_leap_year_method2(year):
    return year % 4 == 0 and (year % 100 != 0 or year % 400 == 0)
# Example usage: year = 2021      if is_leap_year_method2(year):
    print(f"{year} is a leap year.")
else:     print(f"{year} is not a leap
year.") #example usage:

year = 2024 if is_leap_year_method1(year) ==
is_leap_year_method2(year):
    print(f"Both methods agree that {year} is a leap year: {is_leap_year_method1(year)}")
```

output

:

```
PS C:\Users\jeshwanth\.vscode\github> python -u "c:\Users\jeshwanth\.vscode\github\tempCodeRunnerF
2020 is a leap year.
2021 is not a leap year.
Both methods agree that 2024 is a leap year: True
PS C:\Users\jeshwanth\.vscode\github>
```

Task 4: Student Logic (Odd/Even Sum)

Prompt: Write a Python program that calculates the sum of odd and even numbers in a tuple.

Code:

```
def sum_odd_even(numbers):
    odd_sum = sum(x for x in numbers if x % 2 != 0)
even_sum = sum(x for x in numbers if x % 2 == 0)     return
odd_sum, even_sum


# Example usage:
tuple_numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10) odd_total,
even_total = sum_odd_even(tuple_numbers) print(f"Sum
of odd numbers: {odd_total}") print(f"Sum of even
numbers: {even_total}")
```

output

:

```
PS C:\Users\jeshwanth\.vscode\github> python -u "c:\Users\jeshwanth\.vscode\github\tempCodeRunnerF
Sum of odd numbers: 25
Sum of even numbers: 30
PS C:\Users\jeshwanth\.vscode\github>
```